

Gestational Diabetes Mellitus in Relation to Maternal Dietary Heme Iron and Nonheme Iron Intake

.

OBJECTIVE Higher heme iron intake is associated with increased type 2 diabetes risk. However, no previous study has evaluated gestational diabetes mellitus (GDM) risk in relation to heme iron intake during pregnancy. We investigated associations of maternal preconceptional and early pregnancy heme and nonheme iron intake with subsequent GDM risk. **RESEARCH DESIGN AND METHODS** We conducted a prospective cohort study of 3,158 pregnant women. A food frequency questionnaire was used to assess maternal diet. Multivariable generalized linear regression models were used to derive estimates of relative risks (RRs) and 95% CIs. **RESULTS** Approximately 5.0% of the cohort developed GDM ($n = 158$). Heme iron intake was positively and significantly associated with GDM risk ($P_{\text{trend}} = 0.04$). After adjusting for confounders, women reporting the highest heme iron intake levels (≥ 1.52 vs. < 0.48 mg per day) experienced a 3.31-fold-increased GDM risk (95% CI 1.02–10.72). In fully adjusted models, we noted that a 1-mg per day increase in heme iron was associated with a 51% increased GDM risk (RR 1.51 [95% CI 0.99–2.36]). Nonheme iron was inversely, though not statistically significantly, associated with GDM risk, and the corresponding RRs were 1.00, 0.83, 0.62, and 0.61 across quartiles of nonheme iron intake ($P_{\text{trend}} = 0.08$). **CONCLUSIONS** High levels of dietary heme iron intake during the preconceptional and early pregnancy period may be associated with increased GDM risk. Associations of GDM risk with dietary nonheme iron intake are less clear. Confirmation of these findings by future studies is warranted.